- Quality benefits
- Team benefits

2.3.1 Cost saving benefits

Because contract testing sits above unit testing, as represented in our revised test automation pyramid, it offers quicker execution times compared to integration and E2E tests. This means that it can provide additional cost-saving benefits in finding bugs earlier in the software development life cycle. By catching bugs at an early stage, organizations can significantly reduce the time and resources spent on subsequent debugging and fixing. Contract testing takes time, however, because of its higher complexity in the initial setup, which we'll cover as part of section 2.5.1.

An additional cost-saving aspect of contract testing lies in its close integration with the application codebase. Typically, contract tests are co-located within the same codebase as the application itself. This proximity allows developers to easily execute and validate the contract tests locally on their machines before pushing their changes to an environment. By identifying and addressing problems early on, developers can prevent bugs from propagating to production environments. This localized bug fixing not only saves time but also minimizes the effect on the overall build pipelines. The developer experience is also better because they don't need to check out a different project repository or use an unfamiliar tool because there are contract testing tools that allow them to write the tests in their programming language of choice, which is also similar to the codebase language.

We've mentioned that the cost of fixing a bug increases exponentially when it's detected later in the development process. Early bug detection through contract testing helps mitigate costs by identifying problems earlier. By catching bugs before they spread to integrated environments or production systems, organizations can minimize the ripple effects and avoid costly ramifications that may arise.